

Low voltage Circuit Breaker

Lecture 9

Ground Fault Circuit Interrupter (GFCI)

- A [Ground Fault Circuit Interrupter \(GFCI\)](#) is a type of circuit breaker that protects against electric shock by quickly shutting off power when it detects an imbalance in the electrical current. This imbalance typically occurs when current is leaking somewhere, such as through water or a person. GFCIs are commonly used in areas where water and electricity are close together, like bathrooms, kitchens, and outdoor outlets.
- GFI (Ground Fault Interrupter) is a similar term that is sometimes used interchangeably with GFCI. However, technically, GFI is a type of outlet that provides similar protection as a GFCI but does not offer the same level of overall circuit protection as a GFCI circuit breaker does for an entire circuit.

Arc Fault Circuit Interrupter (AFCI) Circuit Breaker

- **AFCI stands for Arc Fault Circuit interrupter**; it is another type of low voltage circuit breaker that offers protection **against Arc faults**. An **arc fault is a high power discharge between two broken conductors**. These arcs can generate enough heat to start fire and cause serious damage to property and life. **The AFCI breaks the circuit upon sensing any arc in the connected circuit.**
- The arcs are generated due to **incorrect or loose joints** in cables or damaged cables by nails, twisting, kinking, etc. **Any loose connection in a PowerPoint or old cables may also cause an electrical arc. It can start fire and disrupt and damage any sensitive electronic equipment.**

The arcs generate a non-periodic waveform that is detected by using a sensitive logic circuit. It discriminates between a normal arc (occur during switching) and an arc fault. As soon as the arc is detected the circuitry trips the power supply but it cannot prevent the first arc. Although it can prevent the ones that follow and avoid a potential fire hazard.

Common Trip (Ganged) Circuit Breakers

- A common trip or Ganged circuit breaker is an assembly of two or more than two circuit breakers that simultaneously break multiple circuits from a fault in only one circuit. The external operating handle of the individual breakers is tied together. It is usually used in a 3-phase system where a fault in a single phase must break the power supply to all the phases. It is also used for increasing the current ratings of a CB by connecting it in parallel.

Magnetic Circuit Breakers

- Another type of voltage breakers that are commonly used nowadays are the magnetic circuit breakers. They are also used for protecting the electrical equipment in case the circuit suffers any problem and an excessively large amount of current flows through it which is dangerous for the circuitry ahead. Loose connections or contacts, wrong wiring or a heavy load attached to the circuit can cause such problems, and as a result of all these, a current overflow occurs in the path.

Coming to the working principle of magnetic circuit breakers, we first see that what is different about them that outstands them from the other breakers. **So the difference lies in the actuating signal.** The actuating signal is the signal which actuates or stimulates the contacts of the breaker, **giving them the notion to maintain or break their contacts depending on the value of the current flowing through the circuit.** So in case of a magnetic circuit breaker, **a solenoid or an electromagnet is attached to the trip unit.** The function of the solenoid or the trip unit is to **generate a magnetic field.** This is done **when the current passes through the solenoid, magnetism develops in it due to changing magnetic flux, and a magnetic field is created around the solenoid.**

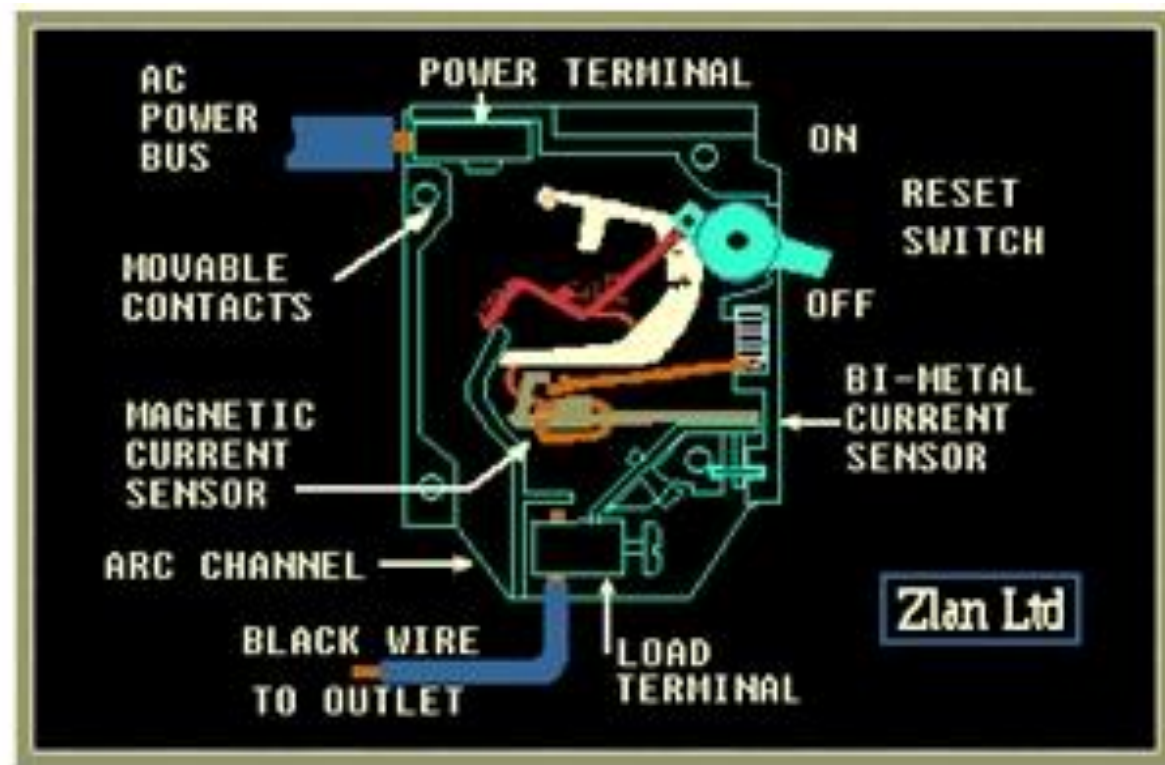
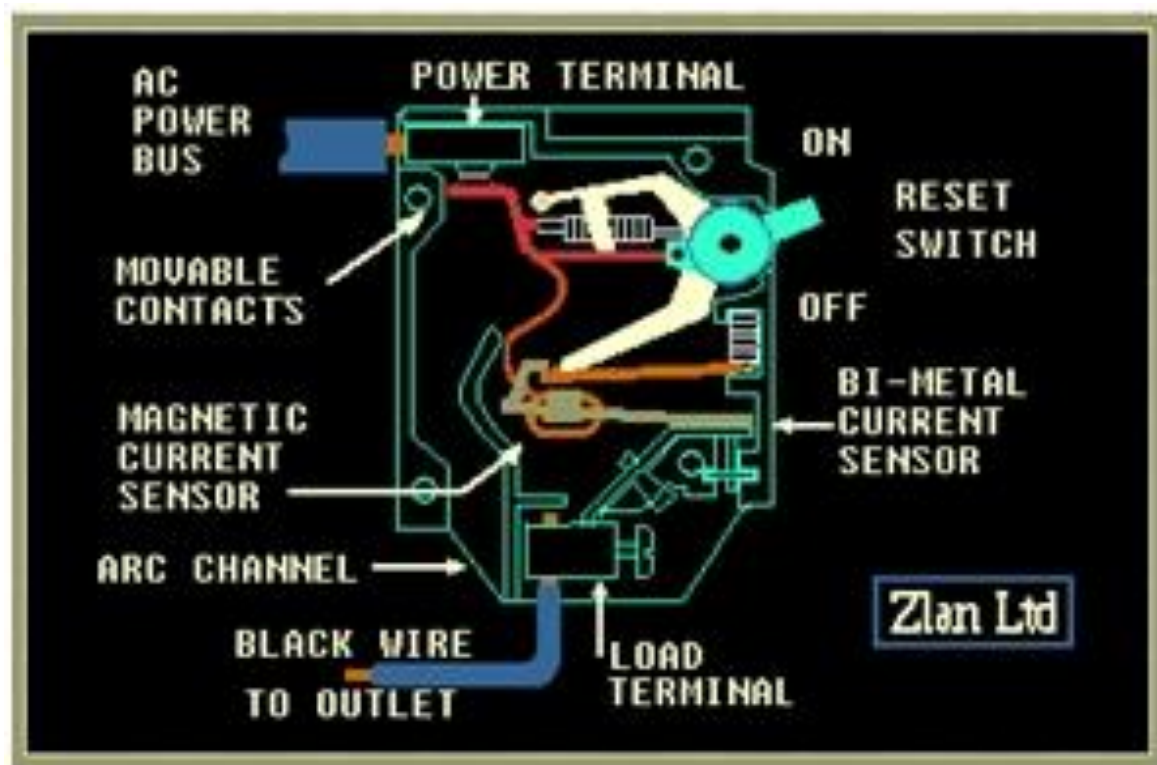
As the strength of this magnetic field is directly proportional to the amount of current applied, so when the current through the circuit increases, the magnetic field around the solenoid or the electromagnet also becomes strong. In case an excessively large amount of current is trying to pass due to some fault occurring in the circuit, the magnetic field becomes so strong that it pulls the bimetallic contact strip so that the contact breaks and the circuit opens and no current can pass any further. When the original amount of the current is stored, i.e. the current is in the safe limits now the magnetic field also weakens a bit. This weak magnetic field does not have enough strength to keep the contact open or pulled towards itself, and hence the contact is restored and immediately the current starts passing through the circuit as before. In this way the circuit breaker operates.

Magnetic Protection

Magnetic protection works much faster than thermal protection. It protects against short circuits, which cause very high currents. Inside the breaker, there is a magnetic coil near the main contacts. When a short circuit happens, the big current makes a strong magnetic field in the coil. This field pulls a plunger or lever and trips the breaker almost right away.

The steps are:

- 1.The circuit works normally. The coil does not move.
- 2.A short circuit makes the current jump up fast.
- 3.The jump makes a strong magnetic field in the coil.
- 4.The field pulls a plunger and opens the latch.
- 5.The breaker trips in less than 10 milliseconds and stops the current.



Thermal Magnetic Miniature Circuit Breaker

- A [thermal magnetic miniature circuit breaker](#) is a safety device. It keeps electrical circuits from getting damaged. **This device has two ways to protect things. The thermal part reacts to slow, steady rises in current. The magnetic part acts fast when there is a sudden surge.** It helps stop wires and equipment from getting too hot or shorting out. People use this device in homes, schools, offices, and factories.
- The main job of a thermal magnetic miniature circuit breaker is to stop electricity if there is a problem. It works very quickly, faster than a blink. After it trips, you can reset it by flipping a switch. This makes it simple to use and helps the electrical devices to run again fast.

A thermal magnetic miniature circuit breaker has many important parts inside. Each part does a special job. The [main parts](#) are:

- Bimetallic Strip:** This strip bends when it gets hot from too much current. If the current stays high, the strip bends more and trips the breaker.
- Magnetic Coil (Solenoid):** This coil makes a strong magnetic field during a short circuit. The field pulls a plunger that trips the breaker right away.
- Latch Mechanism:** This part keeps the contacts closed when things are normal. When the bimetallic strip or magnetic coil moves, the latch lets go and opens the contacts.
- Arc Chute:** When the contacts open, an electric arc forms. The arc chute breaks the arc into smaller pieces and cools it down, making it safe to stop the current.
- Manual Switch:** This lets a person turn the breaker on or off by hand.
- Shell:** The outside case keeps all the parts safe and separates them for each job.

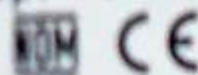
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Surge Protective
Device T2



SUP1-40

Uc: ~220V
Imax: 40kA
In: 20kA
Up: $\leq 1.8\text{kV}$



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Thermal Protection

Thermal protection works by feeling heat from too much current. Inside the breaker, there is a [bimetallic strip](#). This strip has two metals that grow at different speeds when hot. **If the current stays high for a long time, the strip gets hot and bends.** When it bends enough, it opens a latch and stops the electricity.

The steps are:

- 1.The circuit has normal current. The strip stays straight.
- 2.Too much current flows. The strip gets hot.
- 3.The heat makes the strip bend slowly.
- 4.When it bends enough, it lets go of the latch.
- 5.The breaker trips and stops the current.